

ECO 362: Financial Investments

Derivatives

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Derivatives

- Financial contracts whose value depends on an underlying asset.
- Zero net supply: Every long position (buyer) is matched with a short position (seller).
- Zero-sum game: Profits of long and short sum to zero.
- Bad reputation because of the 2008 global financial crisis.
 - “Derivatives are financial weapons of mass destruction” (Warren Buffett).

Types of derivatives

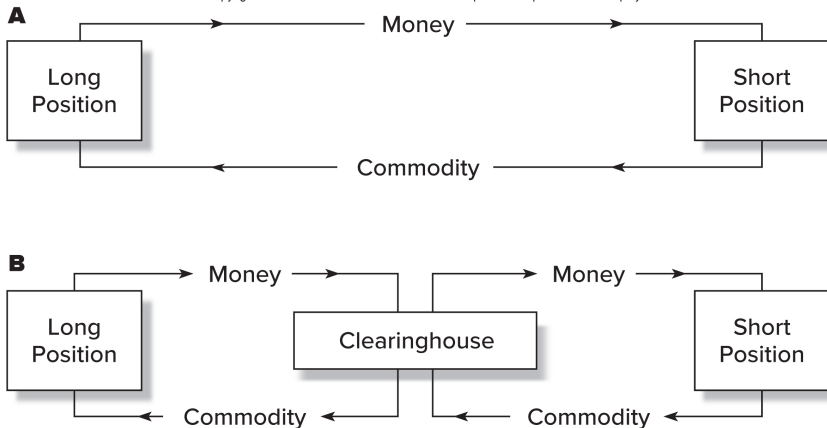
- Types of instruments.
 - Forwards and futures.
 - Swaps.
 - Options.
- Underlying asset.
 - Interest rates.
 - Foreign exchange.
 - Equity.
 - Commodities.
- Over-the-counter (OTC) vs. exchange-traded.

OTC vs. exchange-traded derivatives

- OTC derivatives.
 - Custom contracts.
 - Bilateral trading subject to counterparty risk.
 - Non-transparent prices.
- Exchange-traded derivatives.
 - Standardized contracts.
 - Traded through a central clearinghouse.
 - Margin accounts to minimize default risk.
 - Transparent prices.

OTC vs. exchange-traded derivatives

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Source: Bodie, Kane, and Marcus (2018, Figure 22.3)

Global OTC derivatives

Type	Notional amount (trillion \$)	Market value (trillion \$)
Interest rates	573.697	14.389
Foreign exchange	120.250	4.311
Equity	7.838	0.570
Commodities	2.244	0.301
Credit default swaps	9.902	0.168

Source: Bank for International Settlements (3/2023)

Global exchange-traded derivatives

Instrument	Open interest (trillion \$)
Interest rate futures	36.802
Interest rate options	51.388
Foreign exchange futures	0.330
Foreign exchange options	0.119

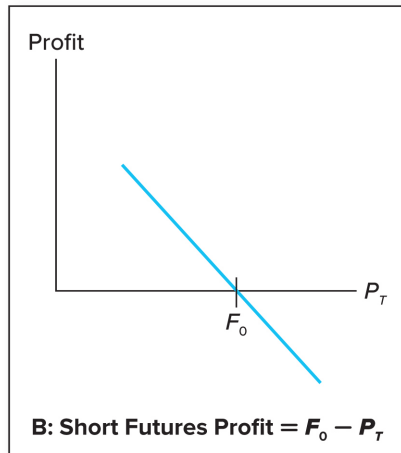
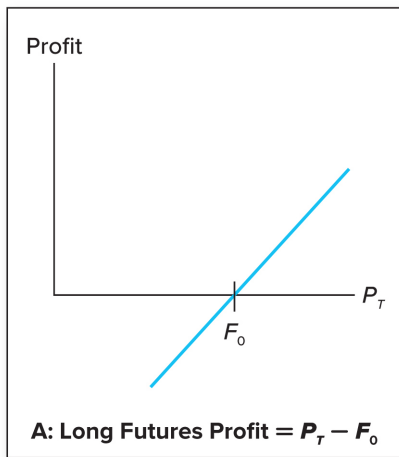
Source: Bank for International Settlements (6/2023)

Futures contract

- A contract to buy/sell Q units of an asset at a predetermined price F in T years from now.
 - F is the futures price.
 - T is maturity.
- Spot price S_0 is the current price of the asset.
- Notional amount is $S_0 \times Q$.
- Cash settlement. If the spot price is S_T at maturity,
 - Buyer's (long) payoff is $(S_T - F)Q$. Profits when spot price increases.
 - Seller's (short) payoff is $(F - S_T)Q$. Profits when spot price decreases.
- Futures contracts are exchange-traded while forward contracts are OTC.

Payoff of a futures contract

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Source: Bodie, Kane, and Marcus (2018, Figure 22.2)

An example of a futures contract

- Current crude oil price is \$50 a barrel.
- Futures price is \$52 at maturity of 6 months.
 - United Airlines goes long 10,000 barrels.
 - Exxon goes short 10,000 barrels.
- Notional amount is \$500,000.
- Why enter a futures contract?
 - United needs to buy fuel but does not want to face uncertainty over future fuel costs.
 - Exxon needs to sell oil but does not want to face uncertainty over future profits.
- The counterparty to United or Exxon could be a speculator (e.g., hedge fund) instead.
 - Long futures to bet price goes up.
 - Short futures to bet price goes down.

Types of futures contracts

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Foreign Currencies	Agricultural	Metals and Energy	Interest Rate Futures	Equity Indexes
British pound	Corn	Copper	Eurodollar	S&P 500 index
Canadian dollar	Oats	Aluminum	Euroyen	Dow Jones Industrials
Japanese yen	Soybeans	Gold	Euro-denominated bond	S&P Midcap 400
Euro	Soybean meal	Platinum	Euroswiss	NASDAQ 100
Swiss franc	Soybean oil	Palladium	Sterling	NYSE index
Australian dollar	Wheat	Silver	British government bond	Russell 2000 index
Mexican peso	Barley	Crude oil	German government bond	Nikkei 225 (Japanese)
Brazilian real	Flaxseed	Heating oil	Italian government bond	FTSE index (British)
	Palm oil	Gas oil	Canadian government bond	CAC-40 (French)
	Rye	Natural gas	Treasury bonds	DAX-30 (German)
	Cattle	Gasoline	Treasury notes	All ordinary (Australian)
	Hogs	Propane	Treasury bills	Toronto 35 (Canadian)
	Pork bellies	Kerosene	LIBOR	Dow Jones Euro STOXX 50
	Cocoa	Fuel oil	EURIBOR	Industry indexes:
	Coffee	Iron ore	Interest rate swaps	• Banking
	Cotton	Electricity	Federal funds rate	• Telecom
	Milk	Weather	Bankers' acceptance	• Utilities
	Orange juice			• Health care
	Sugar			• Technology
	Lumber			
	Rice			

Table 22.1

Sample of futures contracts

Source: Bodie, Kane, and Marcus (2018, Figure 22.1)

Pricing futures by no arbitrage

- For simplicity, assume that the underlying asset
 - Does not pay dividends.
 - Has no storage costs (e.g., financial asset).
- A trading strategy with zero cost today.
 1. Borrow S_0 at continuously compounded interest r_b .
 2. Buy the asset at price S_0 .
 3. Short futures at futures price F and maturity T .
- Payoff at maturity.
 1. Repay $S_0 \exp(r_b T)$.
 2. Asset is worth S_T .
 3. Futures payoff is $F - S_T$.
- Total payoff at maturity is

$$-S_0 \exp(r_b T) + F$$

- If total payoff is positive, there is arbitrage.
- If total payoff is negative, the opposite strategy earns an arbitrage profit.
 1. Lend S_0 at continuously compounded interest r_b .
 2. Short the asset at price S_0 .
 3. Long futures at futures price F and maturity T .
- The only futures price consistent with no arbitrage is

$$F = S_0 \exp(r_b T)$$

Futures price with dividends

- Assume that the underlying asset has a dividend rate that is known in advance.
- Let d be the continuously compounded dividend rate per dollar of the asset.
- A trading strategy with zero cost today.
 1. Borrow $S_0 \exp(-dT)$ at continuously compounded interest r_b .
 2. Buy $\exp(-dT)$ units of the asset at price $S_0 \exp(-dT)$.
 3. Short futures at futures price F and maturity T .
- Payoff at maturity.
 1. Repay $S_0 \exp((r_b - d)T)$.
 2. Asset is worth S_T .
 3. Futures payoff is $F - S_T$.
- By no arbitrage,

$$F = S_0 \exp((r_b - d)T)$$

Futures price with storage costs

- Suppose that the underlying asset has storage costs (e.g., crude oil).
- Let c be the continuously compounded storage cost per dollar of the asset.
- A trading strategy with zero cost today.
 1. Borrow $S_0 \exp(cT)$ at continuously compounded interest r_b .
 2. Buy $\exp(cT)$ units of the asset at price $S_0 \exp(cT)$.
 3. Short futures at futures price F and maturity T .
- Payoff at maturity.
 1. Repay $S_0 \exp((r_b + c)T)$.
 2. Asset is worth S_T .
 3. Futures payoff is $F - S_T$.
- By no arbitrage,

$$F = S_0 \exp((r_b + c)T)$$

Futures return

- Collateralized long futures strategy.
 1. Long futures at futures price F and maturity T .
 2. Lend $F \exp(-r_b T)$ for T years.
- Payoff at maturity.
 1. Futures payoff $S_T - F$.
 2. Receive payment F .
- Continuously compounded return is

$$\begin{aligned}\exp(r_p T) &= \frac{S_T}{F \exp(-r_b T)} \\ &= \frac{S_T}{S_0} \frac{S_0}{F} \exp(r_b T)\end{aligned}$$

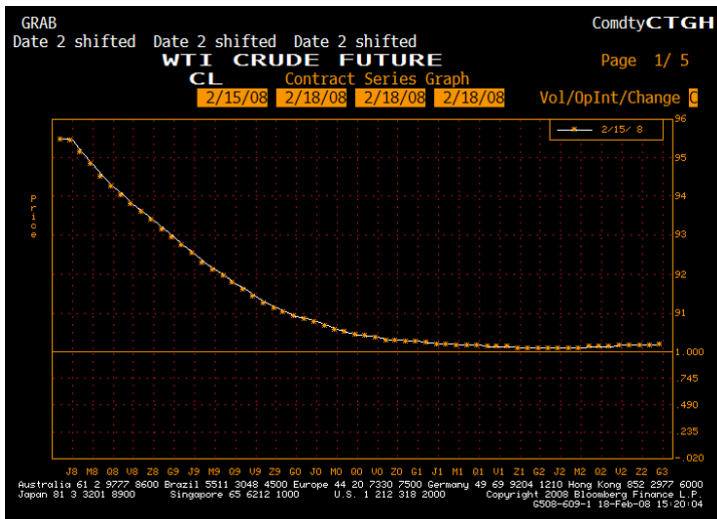
Expected futures return

- Expected excess return is

$$\mathbb{E}[r_p - r_b] = \underbrace{\frac{1}{T} \mathbb{E} \left[\ln \left(\frac{S_T}{S_0} \right) \right]}_{\text{expected spot return}} + \underbrace{\frac{1}{T} \ln \left(\frac{S_0}{F} \right)}_{\text{basis}}$$

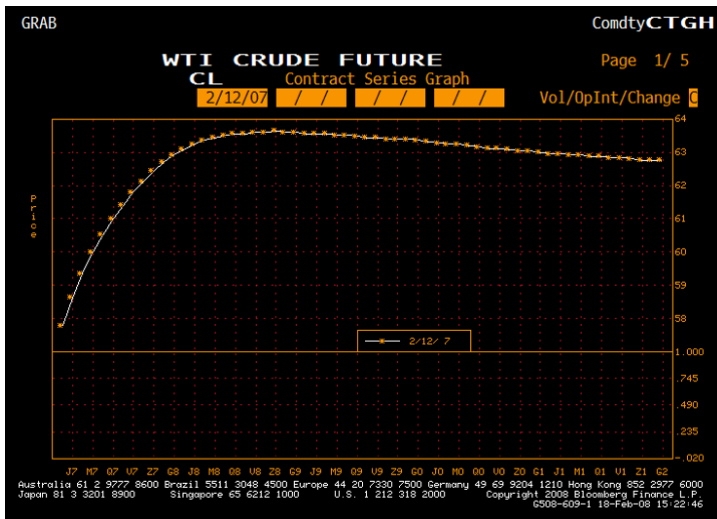
- Expected return depends on basis.
 - Backwardation if $S_0 > F$: Earn a higher return because futures is priced below the spot.
 - Contango if $S_0 < F$: Earn a lower return because futures is priced above the spot.

Crude oil in backwardation (2/15/2008)



Source: Ang (2014, Chapter 11)

Crude oil in contango (2/12/2007)



Source: Ang (2014, Chapter 11)

Interest rate swap

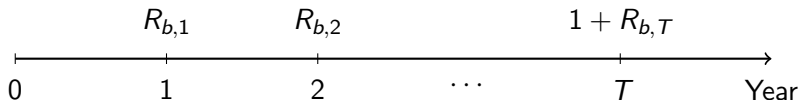
- A contract to exchange a floating rate $R_{b,t}$ for a fixed rate c for T years.
 - $R_{b,t}$ is a short-term interest rate (typically LIBOR).
 - T is maturity (up to 30 years).
 - Payment frequency is monthly, quarterly, semi-annual, or annual.
- Let N be the notional amount.
- On each payment date $t = 1, \dots, T$,
 - Counterparty that receives floating and pays fixed: $(R_{b,t} - c)N$.
 - Counterparty that receives fixed and pays floating: $(c - R_{b,t})N$.
- Set fixed rate c so that the swap has zero value at issuance.

Pricing an interest rate swap by no arbitrage

1. Price a T -year floating rate note with annual coupon $R_{b,t}$ and principal \$1.
2. Price a T -year par bond with coupon rate c and principal \$1.
3. No arbitrage.

Pricing a floating rate note

- T -year floating rate note with annual coupon $R_{b,t}$ and principal \$1.



- Investment strategy that replicates the floating rate note.
 - Year 0: Invest \$1 at interest $R_{b,1}$.
 - Year 1: Take out $R_{b,1}$ from your account and reinvest \$1 at interest $R_{b,2}$.
 - Year 2: Take out $R_{b,2}$ from your account and reinvest \$1 at interest $R_{b,3}$.
 - Year T : Take out $1 + R_{b,T}$ from your account.
- Strategy requires only \$1 initial investment, so the floating rate note must also be worth \$1 by no arbitrage.

Pricing a par bond

- Recall from the lecture on “Discounting” that the par yield at maturity T is the coupon rate c at which a T -year coupon bond trades at face value.

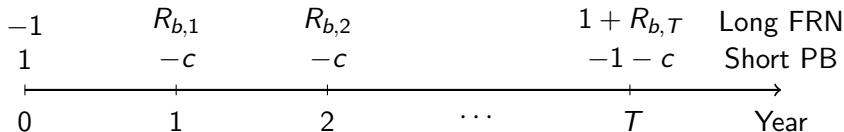
$$1 = \sum_{t=1}^T cP_b(t) + P_b(T)$$

- $P_b(t)$ is the price of t -year zero-coupon bond with \$1 face value.
- Solving for the par yield,

$$c = \frac{1 - P_b(T)}{\sum_{t=1}^T P_b(t)}$$

Pricing an interest rate swap by no arbitrage

- A trading strategy with zero cost today.
 1. Long T -year floating rate note.
 2. Short T -year par bond with par yield c .



- Payoff $R_{b,t} - c$ on each coupon date t .
- Payoffs are identical to an interest rate swap.
- By no arbitrage, the fixed rate c must be the par yield at maturity T .

Hedging through an interest rate swap

- Receiving floating and paying fixed.
 - Equivalent to long floating rate note and short constant coupon bond.
 - Hedge for institutions (e.g., banks) whose assets have fixed interest rates and liabilities have floating interest rates.
- Receiving fixed and paying floating.
 - Equivalent to long constant coupon bond and short floating rate note.
 - Hedge for institutions (e.g., pension funds and insurance companies) whose assets have floating interest rates and liabilities have fixed interest rates.

Speculating through an interest rate swap

- Receiving floating and paying fixed.
 - A bet that interest rates unexpectedly rise.
- Receiving fixed and paying floating.
 - A bet that interest rates unexpectedly fall.
- Interest rate speculation through swaps is risky because of leverage.
 - In 2004, Harvard entered into \$2.3 billion of swaps (receiving floating and paying fixed).
 - Losses amounting to \$910 million during 2008–2012.

Credit default swap

- Insurance on default risk of a corporate or sovereign bond.
- 5-year CDS with spread of 300 bp and notional amount of \$10 million.
 - Buyer makes quarterly payments of

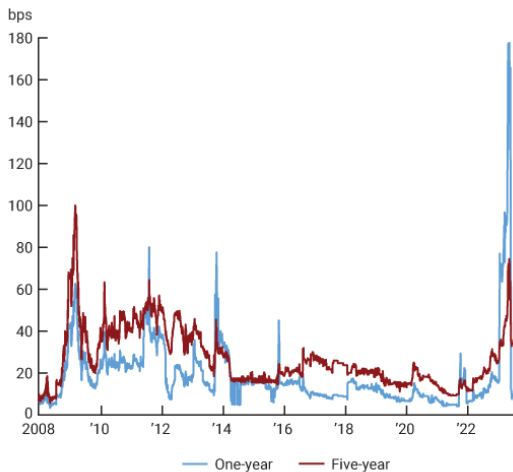
$$\$10 \text{ million} \times \frac{0.03}{4} = \$0.075 \text{ million}$$

- In the event of default, suppose that the recovery rate is 45 cents per dollar of face value. Seller pays

$$\$10 \text{ million} \times (1 - 0.45) = \$5.5 \text{ million}$$

- CDS spread
 - Increases with probability of default.
 - Decreases with expected recovery rate.

US sovereign CDS spread



Source: Benzoni et al. (2023, *Economic Perspectives*)

Summary

- Futures and swaps are useful instruments for hedging interest rate, exchange rate, equity, and commodity price risk.
- Also for speculation, but risky because of leverage.
- Will cover options in next class.